Data Handling and Infrastructure for AI

Data Types

Tamás Grósz

SETU

2026

Data Handling and Infrastructure FOR AI

Learning Objectives

By the end of this lecture, you should be able to:

- Explain why **data is fundamental to AI**.
- Distinguish between major categories of data.
- Recognize common examples of each data type.
- Understand how different data types affect AI systems.
- Identify basic challenges associated with storing and processing data.

Key question

What exactly do we mean when we say “data”?

AI and Data

$$\boxed{\text{Real-world input}} \longrightarrow \boxed{\text{Features } \mathbf{x}} \longrightarrow \boxed{\text{AI model } f_\theta} \longrightarrow \boxed{\text{Prediction } \hat{y}}$$

AI as a mathematical function

At its core, a supervised ML model learns a mathematical function:

$$\hat{y} = f_\theta(\mathbf{x})$$

where $\mathbf{x}$ is the input feature vector and θ represents the learned parameters of the model.

- **Data** provides the examples from which the function is learned.
- **Features** are numerical representations of the information in the input.
- **Training** adjusts the model parameters θ to capture patterns in the data.
- **Inference** applies the learned function to previously unseen inputs.

Why Is Data So Important for AI?

- AI systems learn patterns from data.
- Data determines what a model can learn — and what it cannot learn.
- The quality, quantity, and diversity of data strongly influence model performance.
- Data is needed throughout the AI lifecycle:

Collect $\rightarrow$ **Store** $\rightarrow$ **Clean** $\rightarrow$ **Explore** $\rightarrow$ **Train** $\rightarrow$ **Evaluate** $\rightarrow$ **Deploy**

Important

Building an AI system is often much more about **data and infrastructure** than about choosing the final machine learning algorithm.

The "Garbage In, Garbage Out" Problem

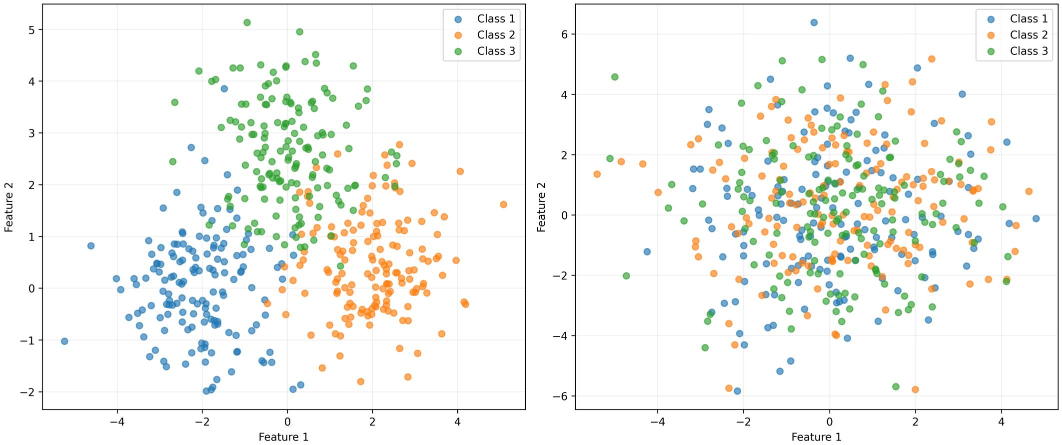
Poor data

- Missing values
- Incorrect labels
- Duplicates
- Bias
- Outliers
- Inconsistent formats

Consequences

- Poor predictions
- Unreliable models
- Hidden biases
- Difficult debugging
- Higher development cost
- Poor generalization

The “Garbage In, Garbage Out” Problem



What Kinds of Data Do AI Systems Use?

A useful first classification is:

1. **Structured data**
2. **Semi-structured data**
3. **Unstructured data**

But we can also classify data by its content:

- Numerical data
- Categorical data
- Text
- Images
- Audio
- Video
- Time-series and sensor data
- Graph/network data

1. Structured Data

Structured data follows a predefined schema.

- Organized into rows and columns.
- Each column has a defined meaning and often a defined type.
- Easy to query and analyze using traditional database tools.

ID	**Age**	**Country**	**Purchase**
001	24	Ireland	125.50
002	37	Germany	82.00
003	29	France	210.75

Examples:

- Customer databases
- Financial transactions
- Hospital records
- Sensor measurements

Numerical Data

Numerical data represents quantities.

Continuous

Values can take many possible values.

- Temperature: $21.7^{\circ} C$
- Height: 1.82 m
- Weight: 74.3 kg

Discrete

Values are typically countable.

- Number of purchases
- Number of users
- Number of defects

Categorical Data

Categorical data represents **groups or labels** rather than quantities.

- Country: Ireland, Germany, Spain, France
- Product type: laptop, phone, tablet
- Customer status: new, returning, inactive
- Animal: cat, dog, horse

Important distinction

Categorical Data usually represented by its index (one-hot encoding).
Categories have meaning, but not necessarily numerical magnitude. ("France" is not three times more than "Germany".)

2. Semi-Structured Data

Semi-structured data does not fit neatly into a table, but still contains organizational information.

Common examples:

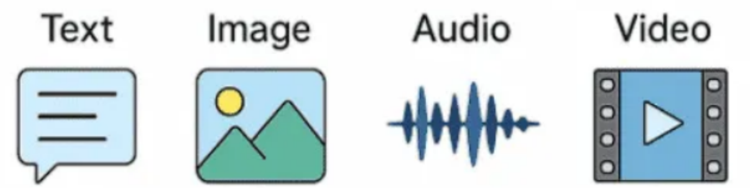
- JSON
- XML
- HTML
- Log files
- API responses

```json
{
  "users": [
    {
      "id": 1,
      "name": "Jane Doe",
      "email": "jane.doe@example.com",
      "isActive": true,
      "roles": ["admin", "user"],
      "profile": {
        "age": 28,
        "location": "Waterford"
      }
    },
    {
      "id": 2,
      "name": "John Smith",
      "email": "john.smith@example.com",
      "isActive": false,
      "roles": ["user"],
      "profile": {
        "age": 34,
        "location": "Dublin"
      }
    }
  ]
}
```

3. Unstructured Data

Unstructured data does not naturally fit into a predefined rows-and-columns schema.

- Images, Audio, Video
- Natural language
- Documents



Why is this important for AI?

Much of the world's useful information is unstructured. AI provides methods for extracting meaningful representations from it.

Text Data

Text is one of the most important data types in modern AI.

Examples:

- Books and articles
- Emails
- Social media
- Customer reviews
- Chat conversations
- Source code

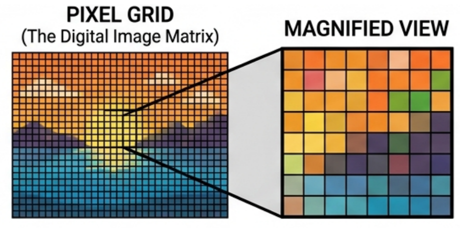
Challenge

Computers do not directly "understand" text. Text must be represented in a form that computational models can process.

Image Data

Images contain information represented by **pixels**.

- Grayscale image: typically one value per pixel.
- Color image: commonly three channels (RGB).
- Image dimensions can range from tiny thumbnails to very high-resolution images.



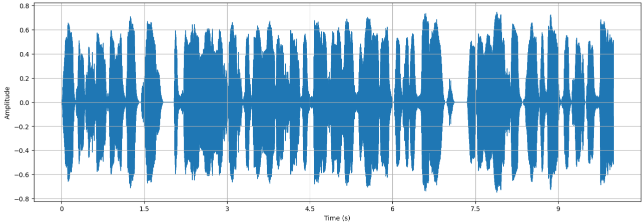
AI examples

Image classification, object detection, facial recognition, medical imaging, autonomous driving.

Audio Data

Audio represents sound as a signal that changes over time.

- Stored as digital samples.
- Sampling rate determines how frequently the signal is measured.



AI examples:

- Speech recognition
- Speaker identification
- Music generation
- Sound event detection

Video Data

Video is essentially a sequence of images over time, often accompanied by audio.

- Frames per second (FPS)
- Resolution
- Compression
- Duration
- Audio streams

AI examples:

- Activity recognition
- Autonomous vehicles
- Security systems
- Sports analytics

Time-Series Data

Time-series data consists of observations associated with a particular point in time.

Examples:

- Stock prices
- Weather measurements
- Electricity consumption
- IoT sensor readings
- Website traffic

Key property

Order matters.

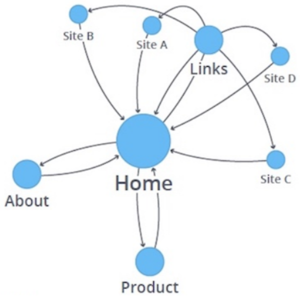
The relationship between observations can depend on time.

Graph and Network Data

Some data is naturally represented as **entities and relationships**.

Examples:

- Social networks
- Transportation networks
- Knowledge graphs
- Recommendation systems
- Molecular structures



Storing graphs

The features of nodes can be stored as structured data
The connections usually stored in an adjacency matrix (often in a sparse matrix)

One AI System Can Use Many Data Types

Consider an **autonomous vehicle**:

- Cameras $\rightarrow$ images / video
- Microphones $\rightarrow$ audio
- GPS $\rightarrow$ geographical coordinates
- Speed sensors $\rightarrow$ time-series data
- Maps $\rightarrow$ structured + graph data
- Vehicle logs $\rightarrow$ semi-structured data

The infrastructure challenge

Different data types have different requirements for **storage, processing, transmission, and analysis**.

Data Type $\neq$ Storage Format

It is important to distinguish between:

What the data is

- Image
- Text
- Number
- Time series
- Graph

How we store it

- CSV
- JSON
- Parquet
- SQL database
- Object storage

Example

A collection of images may be stored as image files in object storage, while metadata about those images may be stored in a SQL database.

Take-Home Messages

1. **Data is the foundation of AI.**
2. Data quality can be as important as model choice.
3. Data can be structured, semi-structured, or unstructured.
4. Important AI data types include:
    - numerical and categorical data
    - text
    - images
    - audio
    - video
    - time series
    - graphs
5. Different data types create different infrastructure challenges.

Discussion: What Data Does AI Need?

Consider an AI system of your choice.

Questions:

1. What types of data does it need?
2. Where could that data come from?
3. How might the data be stored?
4. What problems could occur with the data?
5. What happens if the data is biased or incomplete?

Think about the data before thinking about the model.